

Capturing Data Before Entropy: The Commercial Value of Structured Source Data

Introduction: Unbaking the Data Cake

Enterprise organizations today face a paradox in their information management. On one hand, an estimated **80%–90% of enterprise data is unstructured** ¹ ² – documents, PDFs, slide decks, emails, etc. – and this volume is growing rapidly. On the other hand, **companies are spending billions of dollars** on technologies to make sense of this unstructured data ³. The entire Generative AI and analytics industry is pouring resources into reading and extracting meaning from documents via vector databases, retrieval-augmented generation (RAG) pipelines, large language models (LLMs), and other tools ³. One expert aptly describes this as “*trying to unbake the cake*” – attempting to reverse a process that should never have occurred ³. The uncomfortable truth is that *unstructured* enterprise data is largely an **artificial problem of our own making**. In reality, **data doesn’t start out unstructured – data is born structured**, rich with context, relationships, and meaning ⁴.

Before data gets “baked” into a document, it exists in databases, spreadsheets, forms, sensors, and knowledge graphs. It has **lineage, metadata, and semantic context**. Yet in the final step of publishing information for human consumption, we often **flatten this structured data into static documents**, stripping away its connections and provenance ⁴. We introduce what Timothy Cook calls “*data entropy*” – a loss of order and context that occurs when structured inputs are compressed into a report or PDF ⁴. Ironically, after introducing this entropy, we then **pay handsomely to reverse-engineer** the structure we destroyed, using AI to guess relationships and context that were originally explicit ⁴. In short: *we break the data, then buy expensive glue to fix it*.

This document explores the commercial value of avoiding that entropy in the first place. By capturing and sharing **structured source data** (the “ingredients”) alongside or instead of the final document (“the cake”), organizations can preserve meaning and save enormous effort. We discuss why data born structured should remain so, how much value is lost in the current document-centric paradigm, and how embracing semantic **knowledge graphs** and source-level publishing can unlock economic benefits for both data producers and consumers.

Data Doesn’t Start Unstructured

It’s critical to recognize that what we call “unstructured data” in enterprises is usually a *byproduct of document-centric workflows*, not a naturally occurring phenomenon ⁵. Every glossy 50-page PDF report, slide presentation, or lengthy text document typically **began as structured facts or figures** deep in some system ⁵. For example:

- A quarterly business report may start as rows in a financial database or a spreadsheet.
- An engineering inspection summary might originate from sensor measurements and log records.
- A clinical trial results paper could stem from entries in trial databases or electronic forms.

In each case, the *raw data* has inherent structure (tables, fields, timestamps, relationships). There are connections to source records and contextual metadata (who collected the data, when, how, using what definitions). During analysis and authoring, knowledge workers further organize this information: they form hypotheses, link evidence to conclusions, follow workflows, and often build an internal **ontology or taxonomy** of concepts to make sense of it all. In a sense, **the author or analyst is working with a mental or digital graph of interconnected facts** while developing the content.

However, when it comes time to publish or communicate findings to others, we almost always **“flatten” this rich graph into a linear document**. The structured data (and the author’s mental model of relationships) gets squeezed into paragraphs and bullet points for human-readable form. In that final published document (be it a PDF, DOCX, or printed page), *the explicit links and structure vanish*. Names and terms appear without their full context, references to data lose direct connection to source tables, and conclusions are stated without an easy way to trace back to underlying evidence. In the words of one practitioner, we create static “*photos*” of our data that are **no longer connected to the source materials** ⁶.

All the semantic richness – the **lineage, provenance, and explainability** of how the information was derived – is effectively lost in translation at publish time ⁶. The data hasn’t disappeared, but it has been **“force-compressed” into documents so humans can read it** ⁴. This compression is what introduces *data entropy*: a disordering of information that makes it much harder for machines (or even humans later on) to reassemble the full picture ⁴. In short, **the data starts structured, and becomes unstructured by our own publishing process**.

The High Cost of Reconstructing Lost Structure

Flattening data into documents might make it human-readable, but it offloads a heavy cost to anyone who later needs that data in structured form. Organizations today find themselves doing **extensive post-processing on unstructured data** to get back to the original facts and figures. This happens in a few common ways:

- **Manual Extraction and Re-keying:** Staff may manually read documents and enter key information into spreadsheets or databases for analysis. This is labor-intensive, error-prone, and does not scale.
- **Traditional ETL and Parsing:** Companies use Extract-Transform-Load (ETL) pipelines, custom scripts, or text parsers to pull structured bits out of reports (e.g. using regular expressions or PDF table extractors). This often requires custom code for each document type and still loses nuance that isn’t easily parsed.
- **AI and LLM-Based Extraction:** With the rise of AI, many are now throwing machine learning and LLMs at the problem. Vector databases and LLM-based question-answering systems (RAG pipelines) ingest heaps of PDFs and slides, trying to *infer* the structure and answer questions based on their content ³. These systems essentially treat unstructured documents as a **raw resource to be mined**, guessing at connections after the fact.

Each of these approaches **carries significant cost**. Studies find that a large portion of data professionals’ time (often cited around 70–80%) is spent on data cleaning and preparation rather than actual analysis, due in part to dealing with messy or unstructured inputs. In the context of enterprise documents, the cost is not only in money spent on tools and salaries, but also in *lost accuracy and context*. An algorithm or person trying to interpret a report has none of the original author’s intent or the data’s provenance readily available. They are, in effect, *trying to guess what the author meant*. Important details can be misinterpreted, and biases or assumptions made by the original creator might not be evident, potentially leading to incorrect conclusions.

Timothy Cook summarized this inefficiency bluntly: “We introduce Data Entropy. Then we turn around and pay LLM companies millions of dollars to ‘guess’ the structure we just destroyed. We are breaking the data, and then buying expensive glue to fix it.” ⁷ . In enterprise AI efforts, a **huge portion of the budget is spent trying to reverse-engineer meaning** from unstructured text – essentially attempting to “reverse the soup” or unbake that cake.

Beyond cost, **critical business risks** arise. Without the original context, AI systems might hallucinate or assume wrong context when answering questions. Humans might make decisions on fragmentary data. Moreover, the disconnect from provenance breaks **explainability** – it's hard to trace *why* a piece of data is what it is, or how a conclusion was reached, if you only have the final document. As one expert notes, this process “**breaks Provenance, Explainability and Determinism**” in our data representations ⁶ . In regulated industries or any scenario where auditability and trust matter, this is a serious drawback.

In sum, the current paradigm forces each consumer of information to **reinvent the wheel** by restructuring data from scratch. This one-to-many repetition of effort – many companies and teams all extracting the same report's data separately – is tremendously wasteful. It implies a *large, addressable cost* in the status quo, which savvy organizations could turn into an opportunity.

Why Structured Source Data Is More Valuable than Documents

Given the inefficiencies above, a strong case emerges that the **source structured data (with all its context) is far more valuable than the final document** itself. The document *by itself* is a snapshot, a convenient reading format perhaps, but the **real gold lies in the source materials and their relationships**. Here's why retaining and sharing that structured source (or a machine-readable version of it) has high commercial value:

- **Preserved Context and Provenance:** Source data can include links to original evidence, timestamps, authorship, and the workflow of how insights were derived. This context lets others understand *not just the “what” but the “why”* behind the information. In business terms, it enables better trust and faster onboarding of the data into new analyses, because you don't have to second-guess its meaning or accuracy.
- **Immediate Machine-Readability:** Instead of dealing with PDFs or text, receiving data in a structured form (e.g. a well-defined **knowledge graph, database, or JSON** format) means machines can ingest it directly. This eliminates the costly extraction step. Analysts and AI models can get to work on analysis immediately, accelerating insights and reducing time-to-value.
- **Combining and Reusing Data:** Structured data from multiple sources can be merged or compared much more easily than unstructured documents. For example, if Company A publishes its source dataset (rather than just a report) and Company B does the same, a third party could **join those datasets or run queries across them** to discover new insights. Documents rarely afford this, since they don't *interoperate* – but data does. This **network effect** of data increases its utility and therefore its value.
- **Reduced Errors and Guesswork:** When consumers don't have to interpret free-text, there's less room for misunderstanding. The definitions of fields or relationships in the source data can be clearly specified (through schemas or ontologies). This means decisions made on the data are more likely to be correct, avoiding costly mistakes that might arise from misreading a document.
- **Value of the Full Knowledge Graph:** The source often isn't just rows in a table; it might effectively be a *graph of knowledge*. For instance, an investigative report might implicitly contain a network of people, events, and evidence. The author had that graph in mind while writing. If that graph (the relationships) is captured explicitly in a structured format, it becomes a powerful knowledge asset on its own – one that could drive AI reasoning, recommendation engines, or

complex queries. Many organizations would find such a graph more directly useful than a narrative summary.

From a **commercial standpoint**, if a data producer (say a company publishing a study or a piece of research) can offer the underlying structured data (with rich semantics) for sale or subscription, consumers might gladly pay for it. Why? Because it can **save them significant processing costs** internally. The value of that structured dataset to the consumer is roughly equivalent to the time, money, and inaccuracies they avoid by not having to do document data-mining themselves. As long as the price of the structured data is set *lower than the cost of manual/AI extraction* would be, buying the data is an obvious win. This represents a new **value stream for the producers** of information: they can monetize not just the polished report, but the **foundation of facts and evidence** behind it, which may in fact be *more* useful to many customers.

Moreover, the **combined market** of all the consumers who currently each spend resources parsing the same public document could be substantial. Instead of ten companies each paying (in staff hours or computing costs) to extract the same information from Document B, perhaps those ten could pay the source provider of A a fee, and *all* get the structured A (or a ready-to-use structured version of B). The economics could actually be **favorable for the publisher**, potentially earning them more, while **saving the consumers money overall** (because of eliminating duplicated efforts). In essence, structured data sharing can turn a wasteful one-to-many extraction process into a *one-to-many distribution* of already-structured knowledge.

From Documents Back to Data: A Four-Phase View

To visualize the current situation versus the proposed solution, consider the phases data goes through:

1. **A – Structured Source Data:** This is the beginning – the raw but organized material. It could be a database record, a spreadsheet, or a knowledge graph containing *all the relevant facts, figures, and their interrelationships*. It includes metadata (lineage, author, time, etc.) and possibly an ontology defining how pieces relate.
2. **B – Published Unstructured Document:** This is the typical end product today. *A is transformed into B* for human consumption – for example, an analyst writes a report or an article using data from A. B is usually a **flat, static representation**: sentences, paragraphs, maybe images or tables, but without machine-understandable links to the original source. This phase is a one-to-one transformation (each source dataset yields a document).
3. **C – Reconstructed Structured Data (Intermediate):** Because B is hard for machines to ingest, there may be an intermediate step where someone (the publisher or a third party) *extracts a structured version from B*. Let's call this C, a **machine-readable dataset derived from the document**. In some cases, the publisher might provide this (for example, publishing a CSV or an interactive data dashboard along with a report). If not, consumers might create C themselves via parsing or AI. C aims to represent the content of the document in a structured form (like a table, database, or graph) that computers can work with.
4. **D – Context-Specific Structured Data:** Finally, each consuming organization often takes C (or B directly) and *integrates it into their own context*, yielding D. D is the **company-specific, structured information** ready for that enterprise's internal use. It might be loaded into their data warehouse, combined with their internal data, or annotated with their business-specific taxonomy. This phase is one-to-many: *multiple organizations may independently create their own D from the same B (or C)*.

In the status quo, **A → B is done by the publisher once**, and then **B → D is done by each consumer** (often implicitly going through a C stage via their extraction processes). The pain point is the B → C

conversion: turning an unstructured document back into structured form. This **conversion is currently replicated across many companies**, wasting effort and money repeatedly. It's exactly where the earlier "billions" are spent.

Now imagine a new model: the publisher provides **A (structured data)** in a **consumable format** (with appropriate semantic richness) *either alongside or instead of B*. In an ideal scenario, the publisher might actually maintain a **live knowledge graph or dataset** behind the report. They could publish B for human readers, but also publish the *graph of facts and evidence* that went into B. This would make the **A → C transformation trivial or unnecessary** – essentially C is provided for free as A. Each consumer can then take A and directly produce **their context-specific D with far less effort** (no heavy extraction needed). The overall workflow becomes: A (by publisher) → D (by each consumer with minor adjustments).

The **value proposition** here is clear: the cost that was once incurred in the B → C step (and then into D) is drastically reduced. If Company X would have spent \$100k worth of labor/compute to extract insights from a document, maybe they pay the publisher \$30k for the structured dataset and save \$70k. Multiply that by many consumers, and the publisher can **capture some of the aggregated value** of all those avoided extractions. As long as A is delivered in a standard, *machine-friendly* way, the **time-to-integration** for consumers shrinks, and they can act on the information faster – a competitive advantage.

New Opportunities and Business Models

Embracing structured source data publication opens up several opportunities:

- **Data-as-a-Product Offerings:** Organizations can package their datasets (with proper documentation and ontology) as commercial products or services. Instead of selling just reports or PDF subscriptions, they can sell API access to live data, knowledge graph downloads, or data feeds. This can be a new revenue stream, especially if the data is high-value and others currently spend resources to obtain it indirectly.
- **Collaborative Ecosystems:** If multiple data producers publish structured data, it becomes easier to **mash up data from different sources**. This could give rise to marketplaces or data collaboratives where structured datasets are exchanged, combined, and leveraged, leading to richer insights than any single report would allow. Companies could even form partnerships: e.g., a standards body could define a common schema so that data from various contributors can integrate seamlessly.
- **Innovation in Tools:** The shift to sharing source data will encourage better tools for **graph-based editing, provenance capture, and semantic publishing**. Instead of documents being the final output, documents might become by-products or views of an underlying knowledge graph. We may see authoring tools that capture the *transformation steps* an analyst took (queries run, intermediate calculations, reasoning steps) and bundle that with the final narrative. Such tools would preserve the **transformation of one idea to another** – essentially keeping a log of how raw data was turned into insight.
- **Trust and Transparency as Selling Points:** Companies that provide full context with their data could build **greater trust with their audience and customers**. For instance, a research firm publishing not only conclusions but also the raw data and how it was analyzed is likely to be seen as more transparent and credible. In fields like scientific research, medicine, or finance, this could be a significant differentiator. Consumers might favor data sources that come with provenance and explainability baked in.

- **Reduced Vendor Lock-In:** If the movement is towards open, semantic data formats (like RDF knowledge graphs, JSON-LD, or other open standards), consumers are not locked into specific vendors' extraction pipelines or formats. They can use a variety of tools to work with the data. This openness can accelerate adoption – more companies will be willing to buy into source data if they know it integrates easily with their systems and they won't be stranded if they switch technologies. Using **standard ontologies and taxonomies** to describe data will be key here, ensuring interoperability.

Implementation Considerations

While the benefits are compelling, shifting to a source-data-centric model requires careful thought:

- **Data Governance and Security:** Organizations must decide how to share source data without leaking sensitive information. Not everything can be open. Fine-grained access control, aggregation to protect privacy, or anonymization might be necessary in some cases. The commercial model might involve **licensing agreements** rather than open public release, depending on the data sensitivity.
- **Standardization:** To truly scale this, industries may need to agree on standard schemas or knowledge graph ontologies for common domains. Much like how financial data has XBRL or how schema.org provides web data standards, having common models will make it easier for data from A to plug into a consumer's system with minimal custom work.
- **Metadata Richness:** Publishers should include as much useful metadata as possible – not just the data points, but definitions of fields, units of measurement, timestamps, and links to source evidence. Including **semantic context** (e.g., using Semantic Web technologies) ensures that the meaning of the data travels with the data itself. As noted, capturing the *lineage* and *transformation steps* (if feasible) adds even more value, because a consumer can follow the chain of reasoning.
- **Tools and Training:** Analysts and authors will need tools that let them easily generate these structured outputs. This might mean adopting knowledge-graph-based workflows or augmenting traditional document writing with linked data. Training knowledge workers to think in terms of **graphs and data linking** will help; many may need to learn new skills or use new platforms to author content in this enriched way.
- **Economic Incentives:** To drive adoption, the economics must work out. Producers will ask: does selling the structured data yield enough ROI to justify the extra effort of preparing it? Consumers will ask: do we save enough by buying data vs. extracting it ourselves? Early success stories and pilot projects demonstrating clear ROI will help overcome inertia. In some cases, internal use alone could justify it (even if you're not selling data, capturing it at the source can save *your own* downstream teams time). But the **network effect** truly comes when many organizations participate.

Conclusion: Preventing Unstructured Data at the Source

As the saying goes, *"An ounce of prevention is worth a pound of cure."* Instead of fighting the tide of unstructured data after the fact, enterprises should strive to **prevent data from becoming unstructured in the first place** ⁸. This means reimagining our publication and knowledge-sharing practices. By capturing information in its **native structured state** and sharing those rich, connected data sources, we avoid the costly game of reconstruction later. Tim Cook's advice encapsulates it well: *"If you capture the meaning at the point of creation, you don't need to hallucinate the context later. Stop fixing unstructured data. Start preventing it."* ⁸.

For organizations, the message is clear: there is **commercial value in your data's structure and context**. Treat it as an asset. Whether through internal efficiencies or new data products, maintaining the linkages and provenance of your information can pay dividends. The technology to do this – from knowledge graphs to semantic databases – is maturing, and a community of knowledge graph practitioners is already proving how powerful it can be to have “graphs of graphs” that preserve layered context ⁶. The future of enterprise AI and analytics isn't about ever more clever ways to parse broken data; it's about **not breaking the data to begin with**. By unbaking the cake – or rather, never baking it so fully that ingredients can't be identified – companies can enjoy far richer insights at lower cost. In the end, structured source data isn't just a technical nicety; it's a **strategic resource** that can dramatically streamline information flows and unlock new economic opportunities for all participants in the data value chain.

Sources: The concept of “unbaking the cake” and capturing data before entropy is articulated by Timothy Cook, highlighting the wasted effort in extracting structure from documents ³ ⁴. Dinis Cruz and others emphasize how flattening data into static documents breaks provenance and explainability, advocating for graph-based solutions ⁶. Statistics on the prevalence of unstructured data (80–90% of enterprise data) underscore the scale of the challenge ¹ ². These insights collectively point to the significant commercial value in preserving and sharing structured source data rather than only traditional documents.

¹ ² Unstructured Data: Five Key Statistics You Need to Know

<https://blog.sphereco.com/blog/unstructured-data-5-stats-2>

³ ⁴ ⁵ ⁶ ⁷ ⁸ Addressing Provenance and Explainability in Data Representation | Dinis Cruz posted on the topic | LinkedIn

https://www.linkedin.com/posts/diniscruz_this-is-absolutely-spot-on-and-it-is-crazy-activity-7420815037105172480-huOT